Two Analysts, One Planned Test

Unit 3 · Topic 3.12 · TODAY'S PROBLEM

Suppose an online retailer has 50,000 active accounts. Before collecting data, its team asks, "Does a progress meter change the proportion of active accounts completing checkout?" It randomly selects 360 accounts and randomly assigns 180 to each screen. The table shows the outcomes. Let p_M and p_S be the corresponding population completion proportions.

Checkout outcomes

Screen	n	Complete	Other
Meter	180	126	54
Standard	180	108	72

Nia: "Use $H_0 : p_M - p_S = 0$ versus $H_a : p_M - p_S \neq 0$ because either direction is a change."

Omar: "Because $\hat{p}_M$ is larger, use $H_a : p_M - p_S > 0$."

FIRST TAKE (5 min, on your sheet)

When the team asks whether checkout changes, should a decrease count too? Use the original question to explain.

DOK 1 **(a)** Define p_M and p_S in context. Calculate $\hat{p}_M$, $\hat{p}_S$, and the observed difference $\hat{p}_M - \hat{p}_S$.

DOK 2 **(b)** Name the test procedure. Calculate the pooled proportion and verify randomization, the 10 percent condition, and all four pooled expected counts.

DOK 3 **(c)** In 3–4 sentences, choose and state the hypotheses. Use the original word change and when the direction was chosen. Explain how Omar's choice after seeing the result changes the question and could overstate the evidence.

Video + follow-along: `u6_lesson10_live.html` (scan the code)

Finish (a)–(c) after the video. Turn in your sheet.